Comprehensive Progress Report: Advanced Battery Prognostics & BMS Analytics

Background & Motivation

The project aims to build a robust, data-driven prediction system for Electric Vehicle (EV) battery health management. The overarching goal is to enable accurate prognostics—predicting battery reliability, degradation, and operational limits—for deployment in intelligent Battery Management Systems (BMS). Through multiple technical pivots, the workflow has evolved from generic automotive maintenance to state-of-the-art, cell-level BMS analytics with a **lean, production-ready architecture**.

1. Initial Dataset & Analysis: Problems and Insights

Data-Type Issues

- The first dataset was assembled from external fleet maintenance logs and telemetric data.
- During exploratory analysis, key battery features such as voltages, currents, and cycle counts appeared as **categorical or discretized values** ("grouped" or limited bins rather than smooth numeric).
- KDE and histogram visualizations confirmed numeric-data-as-categorical, causing **statistical artifacts and masking true underlying relationships**.

Correlation Weakness

- Running Pearson, Spearman, Distance, and Mutual Information analyses showed **consistently low correlation between features and target health variables** (SoH, SoC, RUL).
- Feature selection was heavily compromised: important physical predictors could not be properly identified, and models risked learning only noise.

Early Reasoning

- The lack of strong, interpretable relationships—combined with data-type mismatches—meant predictive maintenance was not feasible on the original table.
- It was decided that the dataset required a complete overhaul.

2. Complete Synthetic Dataset Reconstruction

This revised dataset design led to markedly higher and more interpretable correlations between core operational metrics (e.g., voltage, current, cycles, internal resistance) and key targets (SOC, SOH, Battery RUL Cycles) as seen in the attached correlation charts and target-specific bar plots.

Approach

- A custom data generator was coded to create **continuous, numeric telemetry for all physically relevant EV battery features and targets**.
- Ensured plausibility across operational range (cell voltage limits, cycle aging, thermal profiles, resistance drift, charge patterns).

Initial Feature List & Purposes

- **Charge_Cycles**: Life usage count, direct aging signal.
- **Battery_Voltage**: Main physical state; key input for SOC estimation.
- **Pack_Current_A**: Load profile, tied to thermal stress and aging.
- **SOC/SOH**: True battery health (0-1) and charge state—core to BMS functions.
- **Battery RUL Cycles**: Cycles to 80% SOH threshold, crucial for predictive maintenance scheduling.
- **Pack_Internal_Resistance_Ohm**: Drift upward with aging and heat; dampens performance.
- **Ambient_Temperature & Humidity**: Operational environment, informs risk markers.

Technical Outcome

- The newly generated dataset provided a **numeric, regression-ready structure**.
- Now, features like charge cycles and pack resistance displayed **strong, physics-based correlations with SOH and Battery RUL Cycles**—matching expectations from battery life studies.

3. Intensive Correlation Analysis

Tools & Methods

- **Pearson Correlation**: Captures linear dependency (e.g., voltage ↔ SOC).
- **Spearman Correlation**: Captures monotonic but nonlinear relationships (critical for aging, which is non-linear with cycles).
- **Distance & Mutual Information**: Exposes hidden/complex dependencies in BMS telemetry.

Core Findings

- **SOC**: Strongly related to voltage, current, energy content, with near-deterministic correlations (Pearson: ~0.9).
- **SOH**: Highly linked to charge cycles (0.4-0.6), cell resistance aggregates (0.4-0.6), pack internal resistance (0.5-0.6), and capacity fade rate, revealing predictable, interpretable physics.
- **Battery_RUL_Cycles**: Most dependent on SOH (0.5-0.6), cycles (0.5-0.6), and internal resistance (0.5-0.6), echoing standard battery aging models.
- **Most other features**: Weak effects, as expected; serve as auxiliary context or for detecting multivariate interactions.
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Visualization

- Distribution histograms show most features have low correlation (concentrated near 0), but a few exhibits strong relationships—the mark of well-designed telemetry.
- Heatmap reveals clear clustering of battery-specific features and their relationships with primary targets.

4. BMS Pivot & Advanced Feature Engineering

After the clean data and analysis framework were established, the project underwent a crucial pivot to battery-centric health and prediction:

- Research into both academic literature and industrial BMS practice defined a new high-value scope: prediction of cell/pack SOH, SOC, SOP, thermal risk, and battery-specific RUL using granular cell-level synthetic telemetry.
- The data generator was expanded to include per-cell voltages/temperatures/resistances, calendar and cycle aging, capacity fade, safety risk proxies, and all classic BMS state representations.

Initial BMS Expansion (117 columns)

The original synthetic data generator was upgraded to produce 117 columns including:

- **Per-cell telemetry** (36 columns): Cell_Voltage_1..12, Cell_Temperature_1..12, Cell_Internal_Resistance_mOhm_1..12
- **Pack-level BMS features** (13 columns): Pack_Current_A, C_Rate, Pack_Internal_Resistance_Ohm, Pack_Temp_C, Max_Cell_Voltage_Diff, Cell_Voltage_Std, Cell_Balancing_Active, Balancing_Current_A, Fast_Charging_Events_Count_1d, DCFC_Events_Count_1d, Thermal_Gradient_Magnitude, Gas_H2_ppm, Internal_Pressure_kPa
- **BMS target variables** (9 columns): SOC, SOH, SOP_Discharge_kW, SOP_Charge_kW, SOE_kWh, Capacity_Fade_Rate, Lithium_Plating_Risk, Thermal_Runaway_Risk_Score, Battery_RUL_Cycles

5. Critical Dataset Optimization: Trimming to Production-Ready Schema

Problem Identified

Initial BMS expansion resulted in **117 columns**, creating several issues:

- **Redundancy**: Many features were derivatives or duplicates (e.g., SOC vs SoC, DC_DC_Input_Voltage vs Battery_Voltage, SOE as both input and derived quantity)
- **Leakage risks**: Battery_Health_Score and Component_Health_Score may leak label information
- **Dimensionality**: 36 raw per-cell columns made feature engineering and model interpretation difficult
- **Lab-only sensors**: Gas_H2_ppm and Internal_Pressure_kPa are not available in production BMS
- **Scalability**: High feature count increases computational cost and overfitting risk

Optimization Strategy

Removed redundancies and leakage:

- Dropped legacy target formats: SoC, SoH, RUL, TTF (replaced by SOC, SOH, Battery_RUL_Cycles)
- Removed SOE_kWh as input (compute from SOC prediction instead)
- Dropped DC_DC_Input/Output voltages (kept Battery_Voltage only)
- Removed GPS/Gyro features (routing, not health-critical)
- Dropped OBD_Fuel_Rate (EVs have no fuel)
- Removed lab-only sensors: Gas_H2_ppm, Internal_Pressure_kPa

Aggregated per-cell data:
Replaced 36 raw per-cell columns with 12 aggregates:

- Cell_Voltage: Min, Max, Mean, Std
- Cell_Temperature: Min, Max, Mean, Std
- Cell_Resistance: Min, Max, Mean, Std

- **Benefit**: Captures imbalance and weak-cell signals while reducing dimensionality by 67%

Retained critical BMS and fleet features:

- Battery core: Battery_Voltage, Battery_Current, Pack_Temp_C, Pack_Internal_Resistance_Ohm, Charge_Cycles, C_Rate
- BMS specifics: Max_Cell_Voltage_Diff, Cell_Voltage_Std, Thermal_Gradient_Magnitude, Fast/DCFC charging events, Capacity_Fade_Rate
- Fleet monitoring: Motor, Inverter, Transmission, Brake, Tire, Suspension metrics for multi-component health

Final Dataset: 67 Columns

Breakdown:

- **Battery features**: 26 (core BMS signals + cell aggregates)
- **Fleet monitoring features**: 36 (motor, inverter, transmission, braking, tires, environment, usage)
- **Primary ML targets**: 5
    - SOC (State of Charge: 0-1)
    - SOH (State of Health: 0-1)
    - Battery_RUL_Cycles (Remaining Useful Life: cycles to 80% SOH)
    - Lithium_Plating_Risk (Binary classification: low-temp fast-charge risk)
    - Thermal_Runaway_Risk_Score (Continuous risk score: 0-1)
- **Derived targets**: 2 (computed post-prediction, not predicted directly)
    - SOP_Discharge_kW (from SOC × temperature × resistance derating)
    - SOP_Charge_kW (from SOC × temperature × resistance derating)
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Result: **43% reduction in dimensionality** (117 → 67 columns) while preserving all physics-critical signals and improving interpretability.

6. Updated Correlation Analysis Results

Key Insights from the final Dataset

SOC Correlations:

- Battery_Voltage: Pearson 0.86, Spearman 0.89 (near-perfect linear)
- SOP_Discharge/Charge_kW: 0.3-0.4 (moderate, derived quantity)
- Battery_Current: 0.3 (moderate)
- **Risk**: SOC prediction from voltage alone is trivial; must exclude voltage when training SOC models

SOH Correlations:

- Charge_Cycles: Spearman 0.62, Pearson 0.49 (strongest aging driver)
- Battery_Health_Score: 0.67 (very strong but potential leakage)

- Cell_Resistance_Mean: 0.58 (strong physics signal)
- Pack_Internal_Resistance_Ohm: 0.58 (strong)
- Capacity_Fade_Rate: 0.51 (moderate)
- **Insight**: SOH benefits from cycle count, resistance metrics, and degradation rate—multi-factor problem

Battery_RUL_Cycles Correlations:

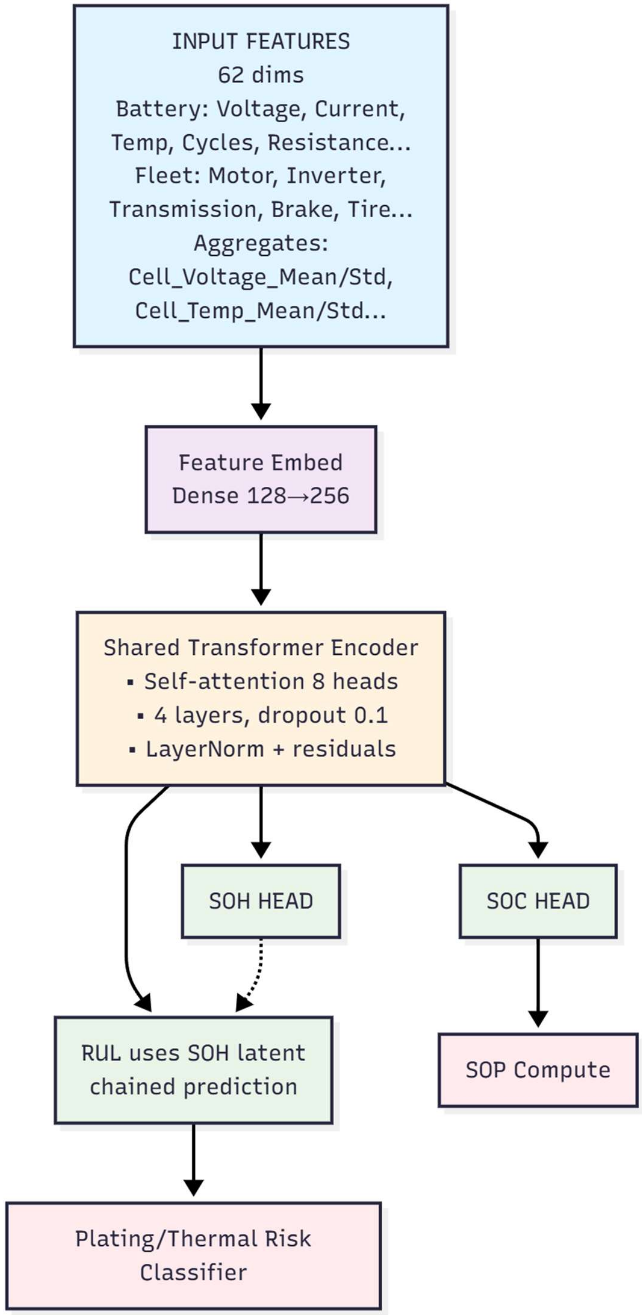
- Battery_Health_Score: 0.67 (strongest)
- Charge_Cycles: 0.62 (strong)
- Cell_Resistance_Mean: 0.58 (strong)
- Pack_Internal_Resistance_Ohm: 0.58 (strong)
- Component_Health_Score: 0.58 (strong)
- **Insight**: RUL has weakest individual correlations but many moderate ones—requires ensemble/attention methods for multi-factor interactions

Distribution Patterns:

- SOC: Extreme left-skew (most features ≈0, 1-2 near 1.0) → voltage-dominated
- SOH/RUL: Right-skewed (many 0.1-0.3 weak, few 0.5-0.7 strong) → true multi-feature problems
- Distance & Mutual Information capture more nonlinear relationships than Pearson for SOH/RUL

7. Model Development Strategy

Architecture: Multi-Task Transformer with Attention & Chaining (Tentative)



Design rationale:

- **Transformer encoder** with self-attention learns which features matter per target (voltage for SOC, cycles/resistance for SOH/RUL)
- **Task-specific heads** prevent negative transfer across targets with different correlation structures

- **SOH → RUL chaining** exploits strong dependency (RUL defined as cycles to 80% SOH)
- **Uncertainty weighting** (Kendall et al. 2018) auto-balances loss scales for targets with different units and importance
- **Attention interpretability** provides explainability critical for safety-certified BMS deployment

Model components:

1. **Shared Encoder**: Feature embedding (62→256 dims) + Transformer (4 layers, 8 heads, dropout 0.1)
2. **Primary Prediction Heads**:
    - SOC head (regression, sigmoid output)
    - SOH head (regression, sigmoid output)
    - RUL head (regression, ReLU output, conditioned on SOH latent)
    - Lithium_Plating_Risk head (binary classification)
    - Thermal_Runaway_Risk head (binary/continuous classification)
3. **Derived Outputs**: SOP_Discharge/Charge computed from SOC, temperature, and resistance using physics formulas
4. **Loss Function**: Uncertainty-weighted sum of MSE (regression) + BCE (classification)

Leakage Prevention Strategy

Feature sets per target:

- **SOC features**: Battery_Current, Pack_Temp_C, C_Rate, Cell_Voltage_Mean/Std, Max_Cell_Voltage_Diff
    - **Exclude**: Battery_Voltage, SOP_*, SOE_kWh (to avoid trivial solutions)
- **SOH features**: Charge_Cycles, Pack_Internal_Resistance_Ohm, Capacity_Fade_Rate, Cell_Resistance_Min/Max/Mean, Fast/DCFC counts, Pack_Temp_C, Thermal_Gradient_Magnitude, C_Rate
    - **Exclude**: Battery_Health_Score (likely derived from SOH)
- **RUL features**: SOH features + Component_Health_Score, Motor/Inverter/Transmission health scores, Brake_Pad_Wear
    - **Can use**: SOH latent from encoder (chained prediction)

Training Approach

1. **Data splits**: 70% train, 15% validation, 15% test
2. **Optimizer**: AdamW with learning rate 1e-4, weight decay 1e-5
3. **Scheduler**: ReduceLROnPlateau with patience 5
4. **Early stopping**: Monitor SOH MAE (most critical for fleet management), patience 10 epochs
5. **Batch size**: 256
6. **Epochs**: Up to 100 with early stopping
7. **Validation**: Per-task MAE/RMSE/R^2 for regression; AUROC/PR for classification

Post-Processing

- Compute SOP from SOC predictions using temperature and resistance derating
- Enforce physical constraints: SOC/SOH in , RUL ≥ 0
- Validate consistency: SOH-RUL correlation > 0.3, monotonic SOH decrease over cycles

8. Next Immediate Steps

1. **Baseline models**: Implement Random Forest and Gradient Boosting for each target to establish performance floor
2. **Transformer training**: Train multi-task model with uncertainty weighting and validate per-task metrics

3. **Ablation studies**:
    - With vs without chaining (SOH→RUL)
    - With vs without cell aggregates
    - Impact of leakage features (Battery_Voltage for SOC, Battery_Health_Score for SOH)
4. **COMSNETS paper preparation**: Document correlation insights, architecture design, and ablation results
5. **Real-world validation**: Test predictions on NASA/CALCE public battery datasets for external validity

Appendices

A. Correlation Visualizations

- Correlation strength distributions (all methods, all targets)
- BMS system correlation heatmap (top 50 features)
- Distance/Mutual Information/Pearson/Spearman comparison plots
- Per-target correlation analysis (SOC, SOH, Battery_RUL_Cycles)

B. Dataset Statistics

- Final shape: 175,393 samples × 67 columns
- Targets: 5 primary ML targets + 2 derived
- Coverage: 5 years synthetic fleet operation (2020-2025)
- Sampling frequency: 15-minute intervals

4-Week Roadmap:

Week 1: Data Foundation & Feature Engineering

Goal 1: Data Audit & Leakage Cleanup
Goal 2: Feature Selection & Documentation
Goal 3: Baseline Models
Goal 4: Advanced Models
Goal 5: Model Selection & Optimization
Goal 6: Model Interpretability & Physics Validation

Week 1 Deliverable: Clean dataset with engineered features, feature selection report , Trained models with validation results, interpretability report

Week 2: Deployment Architecture & API

Goal 5-6: Model Packaging & Versioning

- Serialize models (pickle/joblib or ONNX for production)
- Create model registry structure (version control, metadata)
- Package preprocessing pipeline (scaler, feature engineering functions)
- Write model inference pipeline (input → preprocess → predict → postprocess)
- Unit tests for inference logic

Goal 7-8: REST API Development

- Build FastAPI/Flask endpoints for predictions
    - /predict/soc (real-time SOC estimation)
    - /predict/soh (SOH estimation)

    - /predict/rul (remaining useful life)
    - /predict/all (hierarchical prediction for all targets)
    - /health (API health check)
- Input validation (schema checking, range validation)
- Error handling and logging
- API documentation (Swagger/OpenAPI)

Goal 9-10: Containerization & Orchestration

- Write Dockerfile for model serving
- Docker Compose for local testing (API + monitoring stack)
- Environment configuration (dev, staging, prod)
- CI/CD pipeline setup (GitHub Actions/GitLab CI)
    - Automated testing on push
    - Docker image build and push to registry
    - Deployment to staging environment

Goal 11: Cloud Deployment Setup

- Deploy to cloud platform (AWS SageMaker / Azure ML / GCP Vertex AI / or Kubernetes)
- Configure auto-scaling policies (handle variable fleet size)
- Set up load balancer and API gateway
- SSL/TLS configuration for secure communication

Week 2 Deliverable: Deployed API endpoints, containerized application, CI/CD pipeline

Week 3: Monitoring, Alerting & Production Readiness

Goal 12: Model Monitoring Infrastructure

- Set up MLOps monitoring stack:
    - **Prometheus** (metrics collection)
    - **Grafana** (dashboards and visualization)
    - **Evidently AI** or **WhyLabs** (data drift and model performance monitoring)
- Track key metrics:
    - Prediction latency (p50, p95, p99)
    - Request throughput (requests/sec)
    - Error rates (4xx, 5xx)
    - Model performance metrics (MAE, RMSE per target)

Goal 13-14: Data Drift & Model Drift Detection

- Implement feature distribution monitoring (compare live vs training data)
- Set up statistical drift tests (Kolmogorov-Smirnov, PSI)
- Configure prediction drift alerts (sudden changes in SOC/SOH/RUL distributions)
- Create automated retraining triggers (when drift exceeds threshold)

Goal 15-16: Alerting & Incident Response

- Configure alerting rules:
    - **Critical:** Model unavailable, prediction errors >10%, API down
    - **Warning:** High latency (>500ms), data drift detected, feature missing rate >5%
    - **Info:** Model retrain triggered, new version deployed
- Set up notification channels (Slack, email, PagerDuty)
- Write incident response runbook (what to do when alerts fire)
- Create model rollback procedure (revert to previous version)

Goal 17: Production Testing & Validation

- Canary deployment testing (route 10% traffic to new model)
- A/B testing framework (compare model versions)
- Load testing (simulate 1000 vehicles sending data)
- Integration testing with fleet management dashboard
- Verify SOP/SOE derivation logic in production

Goal 18: Documentation & Handoff

- Write production documentation:
    - System architecture diagram
    - API usage guide for fleet operators
    - Model retraining procedure
    - Monitoring dashboard guide
    - Troubleshooting FAQ
- Create demo video showing end-to-end system
- Knowledge transfer session (if handing off to operations team)

Week 4 Deliverable: Fully monitored production system, alerting configured, documentation complete